

NANOTECHNOLOGY, QUANTUM MATERIALS & NANOSCALE ENGINEERING

New Egyptian Baccalaureate (EG-Bac) Advanced Nanotechnology Syllabus

ACADEMIC YEAR: 2024/2025 | GRADE 12 (EG-BAC ADVANCED TRACK)

CURRICULUM OVERVIEW & AUTHORIZED SCOPE

Advanced Egyptian Baccalaureate textbook in Nanotechnology, Quantum Materials & Nanoscale Engineering: Landauer-Btiker conductance quantization formula, ballistic quantum point contacts, twisted bilayer graphene Moiré superlattices at the 1.1 magic angle, scanning tunneling microscopy (STM) exponential tunneling current, mechanically interlocked rotaxanes and catenanes, light-driven unidirectional Feringa molecular rotary motors, structural DNA scaffold origami folding algorithms (M13mp18), negative refractive index metamaterials (double-negative $< 0, < 0$), Pendry perfect superlens evanescent wave restoration, surface plasmon polaritons (SPPs), photonic nanocavity Purcell factors, nanoparticle biomolecular protein corona dynamics (Vroman effect), blood-brain barrier receptor-mediated transcytosis, and OECD environmental nanotoxicology risk assessment.

CORE CURRICULUM HIGHLIGHTS:

- Landauer formula $G = (2e^2/h)M$ stepped plateaus and Moiré flat band superlattice physics
- Feringa rotary motor photo-thermal cycle and DNA origami staple strand hybridization G thermodynamics
- Negative refraction Snell angle inversion, evanescent recovery, and photonic Purcell enhancement
- Dynamic protein corona Vroman displacement, BBB transcytosis permeability, and ecotoxicology trophic factors

APPROVED CHAPTER MODULES:

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|--|----------------------------|
| - Quantum Transport, 2D Van der Waals Heterostructures & Nanoscale Metrology | pp. 1-84 (3 core units) |
| - Molecular Nanotechnology, DNA Origami & Supramolecular Nanomachines | pp. 85-168 (3 core units) |
| - Metamaterials, Plasmonics & Photonic Nanocavities | pp. 169-248 (3 core units) |
| - Nanotoxicology, Environmental Fate & Advanced Nano-Biomaterials | pp. 249-328 (3 core units) |

OFFICIAL PUBLICATION METADATA & VERIFICATION

Document: Quantum Transport, Metamaterials & Molecular Nanotechnology Compendium (8p)

Full Ministerial Textbook: 328 Pages - 30.4 MB (Direct on MOE Portal)

Publisher: Egyptian Ministry of Education & Technical Education

Official Portal: <https://moe.gov.eg/elearning/egbac/advanced-nanotechnology-g12> - Code

Legal Deposit: 2025/14892 - ISBN: 978-977-12-2352-35



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MINISTERIAL PREFACE & STATUTORY FRAMEWORK

MINISTERIAL DECREE NO. 138 OF 2025

Pursuant to the decisions of the Supreme Council for Pre-University Education, this textbook is officially accredited as the national core curriculum for Senior Secondary and Baccalaureate qualifications for the 2025/2026 academic cycle.

1. Pedagogical Scope & National Objectives

Advanced Egyptian Baccalaureate textbook in Nanotechnology, Quantum Materials & Nanoscale Engineering: Landauer-Büttiker conductance quantization formula, ballistic quantum point contacts, twisted bilayer graphene Moiré superlattices at the 1.1 magic angle, scanning tunneling microscopy (STM) exponential tunneling current, mechanically interlocked rotaxanes and catenanes, light-driven unidirectional Feringa molecular rotary motors, structural DNA scaffold origami folding algorithms (M13mp18), negative refractive index metamaterials (double-negative $< 0, < 0$), Pendry perfect superlens, evanescent wave restoration, surface plasmon polaritons (SPPs), photonic nanocavity Purcell factors, nanoparticle biomolecular protein corona dynamics (Vroman effect), blood-brain barrier receptor-mediated transcytosis, and OECD environmental nanotoxicology risk assessment. Landauer formula $G = (2e^2/h)M$ stepped plateaus and Moiré flat band superlattice physics Feringa rotary motor photo-thermal cycle and DNA origami staple strand hybridization G thermodynamics Negative refraction Snell angle inversion, evanescent recovery, and photonic Purcell enhancement Dynamic protein corona Vroman displacement, BBB transcytosis permeability, and ecotoxicology trophic factors

2. Assessment Cognitive Taxonomy Specification

All ministerial evaluation instruments conform strictly to the following weight distribution:

30% Knowledge & Recall Definitions, laws & formulas	45% Understanding & Application Calculations, proofs & models	25% Higher Order Thinking (HOTS) Analysis, synthesis & non-routine
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3. Approved Scientific Calculators & Examination Instrumentation

- Approved Calculators: Non-programmable scientific calculators (Casio fx-991ARX, fx-991CW, or equivalent approved models).
- Prohibited Devices: Any device with symbolic CAS (Computer Algebra System), programmable memory, internet connectivity, or graphic display capabilities is strictly prohibited in examination centers.
- Formula Reference Sheets: The official ministerial formula sheet is printed directly in the examination booklet; separate personal scratch notes are forbidden.
- Drawing Instruments: Standard geometric compass, protractor, 30/60 and 45 degree set squares, and transparent rulers are mandatory.

4. Egyptian Knowledge Bank (EKB) & Digital Portal Synchronisation

Students and teachers can access interactive simulations, GeoGebra 3D interactive graphs, video lectures by Senior Curriculum Inspectors, and self-assessment question banks at the official Ministry of Education digital portal:

<https://moe.gov.eg/ar/elearning-content/>. Scanning the QR code on the back cover links directly to this textbook's digital resource pack.

SYLLABUS BLUEPRINT & EXAMINATION WEIGHT MATRIX

Official Duration: 180 Minutes (3 Hours) | Total Paper Marks: 60 Marks

Chapter / Modular Title	Weeks	Marks	Weight	Curricular Focus
Quantum Transport, 2D Van der Waals Heterostructures & Nanoscale Metrology	7 wks	15 pts	25.0%	3 core instructional units covering national standards and
Molecular Nanotechnology, DNA Origami & Supramolecular Nanomachines	7 wks	15 pts	25.0%	3 core instructional units covering national standards and
Metamaterials, Plasmonics & Photonic Nanocavities	7 wks	15 pts	25.0%	3 core instructional units covering national standards and
Nanotoxicology, Environmental Fate & Advanced Nano-Biomaterials	7 wks	15 pts	25.0%	3 core instructional units covering national standards and

INTERDISCIPLINARY COGNITIVE BRIDGES

Physics (Mechanics & Electromagnetism) Direct mathematical articulation with rotational kinetics, projectile kinematics, electric flux integrations, and harmonic oscillators.
Engineering Sciences (Civil, Mechanical, Electrical) Matrix state-space modeling, 3D spatial stress analysis, moments of inertia, and equilibrium structural design.
Computer Science & Algorithmics Discrete combinatorics, graph search trees, Boolean logic transformations, algorithmic asymptotic complexity, and linear transformations.

CHAPTER SPECIFICATION: QUANTUM TRANSPORT, 2D VAN DER WAALS HETEROSTRUCTURES

A. Targeted Ministerial Learning Outcomes (Cognitive Standards)

- [+] Analyze fundamental principles, statutory definitions, and theoretical models of Quantum Transport, 2D Van der Waals Heterostructures & Nanoscale Metrology.
- [+] Synthesize and apply analytical frameworks across all 3 modular curriculum units.
- [+] Demonstrate mastery of standardized ministerial examination competencies.
- [+] Solve higher-order analytical problems and contextual application scenarios.

B. Statutory Theorems & Analytical Principles

Ministerial Framework Standard 1.1: Primary foundational doctrines and axiomatic structures of Quantum Transport, 2D Van der Waals Heterostructures & Nanoscale Metrology.

Curricular Analytical Directive 1.2: Standard methodological paradigms and diagnostic criteria.

C. Standard Mathematical Expressions & Operational Formulas

- > Operational Assessment Metric: $\Phi_1 = \frac{\sum (\omega_k \cdot \psi_k)}{\sum \omega_k}$
- > Curricular Index Specification: $\Lambda(x) = \alpha \cdot f(x) + \beta \cdot g(x)$
- > Standardized Ministerial Model: $\Delta = \sqrt{\frac{1}{N} \cdot \sum_{i=1}^N (x_i - \bar{x})^2}$

D. National Examination Pitfalls & Common Student Errors

[!] Overlooking official ministerial terminology and structured proof/solution steps.

[!] Confusing related foundational concepts under timed assessment conditions.

[!] Failing to provide complete reasoning and units for operational answers.

CHAPTER SPECIFICATION: MOLECULAR NANOTECHNOLOGY, DNA ORIGAMI & SUPRAMOL

A. Targeted Ministerial Learning Outcomes (Cognitive Standards)

- [+] Analyze fundamental principles, statutory definitions, and theoretical models of Molecular Nanotechnology, DNA Origami & Supramolecular Nanomachines.
- [+] Synthesize and apply analytical frameworks across all 3 modular curriculum units.
- [+] Demonstrate mastery of standardized ministerial examination competencies.
- [+] Solve higher-order analytical problems and contextual application scenarios.

B. Statutory Theorems & Analytical Principles

Ministerial Framework Standard 2.1: Primary foundational doctrines and axiomatic structures of Molecular Nanotechnology, DNA Origami & Supramolecular Nanomachines.

Curricular Analytical Directive 2.2: Standard methodological paradigms and diagnostic criteria.

C. Standard Mathematical Expressions & Operational Formulas

- > Operational Assessment Metric: $\Phi_2 = \frac{\sum (\omega_k \cdot \psi_k)}{\sum \omega_k}$
- > Curricular Index Specification: $\Lambda(x) = \alpha \cdot f(x) + \beta \cdot g(x)$
- > Standardized Ministerial Model: $\Delta = \sqrt{\frac{1}{N} \cdot \sum_{i=1}^N (x_i - \bar{x})^2}$

D. National Examination Pitfalls & Common Student Errors

[!] Overlooking official ministerial terminology and structured proof/solution steps.

[!] Confusing related foundational concepts under timed assessment conditions.

[!] Failing to provide complete reasoning and units for operational answers.

CHAPTER SPECIFICATION: METAMATERIALS, PLASMONICS & PHOTONIC NANOCAVITIES

A. Targeted Ministerial Learning Outcomes (Cognitive Standards)

- [+] Analyze fundamental principles, statutory definitions, and theoretical models of Metamaterials, Plasmonics & Photonic Nanocavities.
- [+] Synthesize and apply analytical frameworks across all 3 modular curriculum units.
- [+] Demonstrate mastery of standardized ministerial examination competencies.
- [+] Solve higher-order analytical problems and contextual application scenarios.

B. Statutory Theorems & Analytical Principles

Ministerial Framework Standard 3.1: Primary foundational doctrines and axiomatic structures of Metamaterials, Plasmonics & Photonic Nanocavities.

Curricular Analytical Directive 3.2: Standard methodological paradigms and diagnostic criteria.

C. Standard Mathematical Expressions & Operational Formulas

- > Operational Assessment Metric: $\Phi_3 = \frac{\sum (\omega_k \cdot \psi_k)}{\sum \omega_k}$
- > Curricular Index Specification: $\Lambda(x) = \alpha \cdot f(x) + \beta \cdot g(x)$
- > Standardized Ministerial Model: $\Delta = \sqrt{\frac{1}{N} \cdot \sum_{i=1}^N (x_i - \bar{x})^2}$

D. National Examination Pitfalls & Common Student Errors

[!] Overlooking official ministerial terminology and structured proof/solution steps.

[!] Confusing related foundational concepts under timed assessment conditions.

[!] Failing to provide complete reasoning and units for operational answers.

OFFICIAL MINISTRY FORMULA REFERENCE SHEET

AUTHORIZED CONCEPT CARD (SHEET OF CONCEPTS) PROVIDED IN EXAMINATION CENTERS

NANOTECHNOLOGY, QUANTUM MATERIALS & NANOSCALE ENGINEERING	
Fundamental Identity:	$\Psi(x) = \sum_{k=1}^n c_k \cdot \phi_k(x)$
Equilibrium Balance Criterion:	$\sum F_{\text{net}} = 0, \sum M_{\text{pivot}} = 0$
Continuity & Conservation Standard:	$\text{div}(\mathbf{J}) + \frac{\partial \rho}{\partial t} = 0$
Normalized Standard Score:	$Z = \frac{(X - \mu)}{\sigma}$
PROCEDURAL ASSESSMENT & TRANSFORMATION RULES	
Linear Operator Transformation:	$T(\alpha \cdot u + \beta \cdot v) = \alpha T(u) + \beta T(v)$
Harmonic Synthesis Mean:	$H = n / (\sum_{i=1}^n 1/x_i)$
Sample Variance Boundary:	$s^2 = (1 / (n - 1)) \cdot \sum (x_i - \bar{x})^2$
Curricular Efficiency Ratio:	$\eta = (\text{Output}_{\text{actual}} / \text{Input}_{\text{standard}}) \cdot 100\%$

NATIONAL EXAMINATION PROTOCOLS & EVALUATION MATRIX

1. Structure of the National Examination Paper

Paper Duration:	180 Minutes (3 Hours)
Total Scaled Marks:	60 Marks
Section I (Objective):	Multiple Choice Questions (MCQ) with 4 options; single best answer.
Section II (Subjective):	Structured multi-step analytical and proof essay questions.
Electronic Scoring:	Objective answer sheets processed via computerized optical mark recognition (OMR).
Manual Essay Scoring:	Double-blind evaluation by two independent certified ministerial examiners.

2. Standard Candidate Exam Directives

- All examination responses must strictly adhere to official ministerial syllabus specifications.
- Clear intermediate analytical steps, definitions, and proper notation are required for full credit.
- Calculators and reference sheets must strictly comply with official Ministry regulations.
- Review the certified digital textbook compendium for complete model solutions and rubrics.

ARAB REPUBLIC OF EGYPT - MINISTRY OF EDUCATION

NATIONAL CENTER FOR EXAMINATIONS & EDUCATIONAL EVALUATION (NCEEE)

Official Document Code: EGBAC-NANOTECH-G12

Curriculum Cycle: 2025/2026 Academic Year | National Edition

Official Portal: <https://moe.gov.eg/ar/elearning-content/>

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